Lecture 6: Energy

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

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1 Lecture 5: Energy, Food & Humanity

2 Final Project

Final project summary is now on the website. A link was sent via email. Please review it.

Initial project proposal due 05/04 – one week from today. Send before next lecture.

3 Comments about energy

Energy – less intuitive, more challenging, but very important concept. Why is it difficult? We **lack intuition** base on direct experience and, also, **energy is hard to measure**.

Why should we care? Thinking about energy lets us reason about the processes that can take place (for which there is enough energy), reason about resources (is there enough food or fuel for X), and learn about priorities (where the bulk of energy spent by societies, ecosystems, organisms, or cells).

4 Energy bar

My friend dared me to join the Empire State Building Run-Up – a race up the building (102 stories, ≈ 1200 ft tall, observation decks 320 m above ground). How much food should I eat/carry?

Another way of asking the same question: **how high can I climb on a single energy bar?** How would you go about this?

Our calculation scheme: weight $\times$ energy density $\rightarrow$ total energy $\rightarrow$ height (via mgh)

How much does an energy bar weigh? A can of tuna weighs $\approx 150g$ and a snac A lb is about half a kg (0.45 kg) and a snack bar is much less than a pound. I checked and Kind bars weigh 40 g (some are 60 g).[1]

How many calories per bar? **Anyone remember anything about the energy density of food?**

Reminder: food is usually denominated in **C**alories.

$$1 \text{ Cal} = 1\text{kCal} = 1000 \text{ calories} = 4184 \text{ J} \approx 4000 \text{ J} = 4000 \text{ kg m}^2 \text{ s}^{-2}$$

$$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$$

Assume the bar is 50 g (more than a snack!) and the calories are mostly carbs/protein (4 kcal/g, see Table 1).

$$4\text{kcal/g} \times 50\text{g} \implies 200 \text{ kcal/bar} \approx 800 \text{ kJ/bar}$$

If our bodies were 100% efficient

$$\Delta E = mgh, \quad m \approx 10^2 \text{ kg}, \quad g = 10 \text{ m/s}^2$$

[1] Ron's notes use 25 g/bar. No surprise that American bars are bigger!

Outline for whiteboard

- HW #2 – due today
- HW #3 – posted today/tomorrow
- Final projects – 1pager on website
- Ambassadors for guest lectures
- Energy bar
- Conservation
- Butter / bread
- Energy content
- Ag (person $\times$ yr)
- Human energy – break DIY
- Units (power/energy)
- Land for food
- Bicycle or car

Food	kcal/g	kJ/g
Fat	≈ 9	≈ 37
Carbs	≈ 4	≈ 16
Protein	≈ 4	≈ 16

Table 1: Energy density of different macro nutrients. Note that these values are estimates based on the Atwater System of accounting for unmetabolized byproducts and estimates vary based on the food, e.g., protein in eggs vs. protein in grains. This chapter from the FAO is useful for building deeper understanding.

SUBSTANCE	COMBUSTION ENERGY	CARBON CONTENT	CO_2 YIELD
glucose	15.5×10^6 J / kg	≈40%	≈10^{-7} kg CO_2 / J
dry wood	(15 – 18) × 10^6 J / kg	≈50%	≈10^{-7} kg CO_2 / J
coal	(15 – 32) × 10^6 J / kg	≈70% –90%	≈ (1–2) × 10^{-7} kg CO_2 / J
natural gas	≈ 45 × 10^6 J / kg	≈75%	≈6 × 10^{-8} kg CO_2 / J
gasoline & diesel fuel	≈ 45 × 10^6 J / kg	≈80%	≈7 × 10^{-8} kg CO_2 / J

Figure 1: The energy densities of common fuels. These are combustion energies, while the energies in Table 1 are metabolic. Notice that glucose is fairly representative of all combustible fuels, at least within a factor of 3. Notice also that fuels with higher energy density are less oxidized (more reduced) and, consequently, have greater carbon content on a mass basis.

$$\implies h \approx \frac{800 \times 10^3 \text{ J}}{100 \text{ kg} \times 10 \text{ m/s}^2} = \frac{800 \times 10^3 \text{ kg m}^2 \text{ s}^{-2}}{10^3 \text{ kg m s}^{-2}} = 800 \text{ m}$$

Of course, our bodies are not perfectly efficient at converting food into mechanical work like climbing. **How much should I discount for metabolic inefficiency?** Textbooks give $\approx 25\%$ efficiency, giving ≈ 200 m climbed bar.

4.1 Another way that's illuminating

Calculate energy / height climbed

$$\frac{\Delta E}{h} = gm \approx 10\text{m}\,\text{s}^{-2} \times 100\text{kg} = 10^3\,\text{kg}\,\text{m}\,\text{s}^{-2} = 10^3 \text{ J/m}$$

because 1 J $= 1\text{kg}\,\text{m}^2\,\text{s}^{-2}$. So we need about 1 kJ $\approx \frac{1}{4}$ kCal to climb 1 meter at perfect efficiency, giving 800 m/bar. At 25% efficiency, this is 1 kCal/m, meaning 200 m/bar.

4.2 Q: can I power an elevator with energy bars?

Can I burn an energy bar and raise myself 800 m in a (100% efficient) elevator? Seems like a lot. But indeed, this is what we got.

5 Energy conservation, dimensions, and biology

We just did a calculation based on energy conservation – comparing energy input to output as mechanical work. Pretty powerful "no go" condition – if you didn't have enough energy, you definitely couldn't climb the Empire State Building. The calculation highlights that **the most important thing about energy is that it is conserved**.

To emphasize this point, let's think about the impact velocity v_{impact} of a ball falling from rest at height h with no air resistance as in Fig. 2. The usual way of getting v_{impact} is via calculus, but we can make a lot of progress with dimensional analysis.

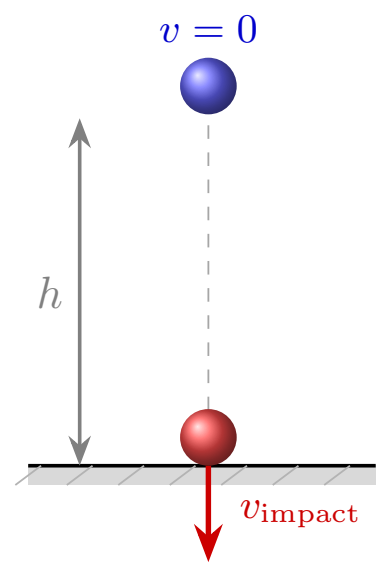

Figure 2: A ball falls from rest at a height h with no air resistance – what is the impact velocity v_{impact}?

We can **approach this question with dimensional analysis**. Acceleration due to gravity, g, has dimensions $[LT^{-2}]$, h has dimensions of $[L]$, and v_{impact} has

dimensions $[LT^{-1}]$. The T^{-1} must come from $\sqrt{g}$, so we infer that[2]

$$v_{impact} \sim \sqrt{gh}$$

We can get an equality with calculus, which gives

$$v_{impact} = \sqrt{2gh}$$

5.1 Falling ball by energy conservation

Energy conservation offers another approach to this problem. You might have seen this in high school physics, but we're going to do it in reverse.

We **postulate a conserved quantity called energy**. We know that when a force acts over a distance it does something to the motion of an object — it changes its energy. So we'll assume that energy has the dimensions of work. In our example, gravity acts on the ball over a distance h and performs work x

$$W = F \cdot h = mgh$$

on the ball (work = force × distance). So energy has dimensions of $[M\ L^2\ T^{-2}]$.

Since we **assumed that energy is conserved** all the work done by gravity must be converted into energy associated with motion ("kinetic energy") at the moment of impact (velocity v_{impact} assuming no air resistance). The dimensions of m and v_{impact} imply that kinetic energy

$$KE \sim mv_{impact}^2$$

By doing some calculus (below), we can find the full expression $KE = \frac{1}{2}mv^2$. Setting kinetic and gravitational energy equal gives

$$mgh = \frac{1}{2}mv_{impact}^2 \implies v_{impact} = \sqrt{2gh}$$

This example emphasizes several important points.

1. **Energy conservation gave the same answer as calculus** but with much less work.
2. If there was air resistance, then $v_{impact} \leq \sqrt{2gh}$ and we have a useful upper bound.
3. More generally, **energy conservation lets us reason about possible outcomes of a process** (impact velocity) **without tracking its trajectory** (the fall).

This is why energy concepts are useful in biology – if we can quantify the available energy, we can infer limits on what its possible for biology to do, even when the underlying details are complicated or unknown.

5.2 Solving for v_{impact} by integration

We can get the constants with calculus.

$$\frac{d^2y}{dt^2} = -g \implies \frac{dy}{dt} = -gt + C_1$$

[2]See Ch. 1 of Street-Fighting Mathematics, which works this example in detail.

Where $C_1 = 0$ because the initial velocity $\left.\frac{dy}{dt}\right|_{t=0} = 0$. Integrating again gives

$$\frac{dy}{dt} = -gt \implies y(t) = -\frac{1}{2}gt^2 + h$$

Solve for the time of impact t_{impact} by setting

$$y(t_{impact}) = 0 = -\frac{1}{2}gt_{impact}^2 + h \implies t_{impact} = \sqrt{2h/g}$$

Now we can solve for v_{impact} as

$$\frac{dy}{dt}(t_{impact}) = -gt_{impact} = \sqrt{2gh}$$

6 Why is butter more energy dense than bread?

You might think that fats have more calories than carbohydrates because they are chemically distinct, or because we metabolize them differently. These are true statements, but not the main reason for the difference in energy density summarized in Table 1.

Our metabolism turns both fats and sugars into CO_2 by breaking bonds and transferring bonding electrons to O_2.

The repeating unit in fats has MW = 14 (Fig. 4) instead of MW = 30 (Fig. 3), giving a factor of $30/14 > 2$, very close to 9/4.

So the primary reason that fats have more calories simply because the building block of fats is more dense.

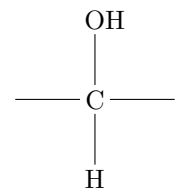

Figure 3: The repeating unit in a lipid has a mass of $2 + 16 + 12 = 30$ g/mol.

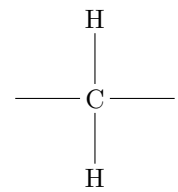

Figure 4: The repeating unit in a lipid has a mass of $2 + 12 = 14$ g/mol.

6.1 Q: Aren't fats also more reduced? i.e., have more electrons to give.

There is also a factor of 6/4 in electrons transferred to O_2:

$$CH_2O + O_2 \rightarrow CO_2 + H_2O \quad (4 \text{ electrons transferred})$$

$$CH_2 + \tfrac{3}{2}\,O_2 \rightarrow CO_2 + H_2O \quad (6 \text{ electrons transferred})$$

In human metabolism, these electrons generally are extracted from the food and donated to O_2. However, not all electron transfers are "worth" the same amount of energy so we shouldn't just multiply by the number of e- transferred.

7 Food energy comes from electron transfer

How much energy is in the food we eat?

Our bodies get energy by respiring (**burning**) food substrates like glucose in O_2.

$$\mathrm{C_6H_{16}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O.}$$

In this reaction, hydrides (H^+ with an electron pair, or H^-) moved from C atoms in glucose to O atoms in O_2, as illustrated in Fig. 5.

As we discussed above, **it does matter what type of food we eat**, but mostly because they have different densities. The fact that they contain different types of bonds (C-H, C-O, C-N, C-C, etc.) matters less.

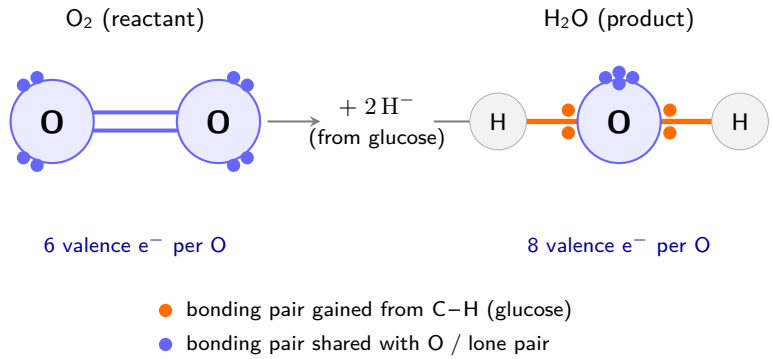

Figure 5: O atoms gain valence electrons during respiration. These valence electrons come from the food. Since glucose has 12 H atoms and CO2 has none, the respiration of glucose involves the transfer of a 12 hydrides (H^-) and, therefore, 24 electrons from glucose to O_2. Diagram made with the help of Claude.

Rule of Thumb: Covalent Bond Energy

Breaking a covalent bond is "worth" about 300 kJ/mol.

Treat sugars as polymers of CH_2O – glucose has 6 of these. Each monomer of food oxidation involves the movement of a few (4) electrons over about 1 volt of redox potential, so a total of a few eV.[3] Because $1\ \text{J} = 1\ \text{C} \times 1\ \text{V}$:

$$\text{f eV} = f \times \frac{1.6 \times 10^{-19}\ \text{J}}{1\ \text{eV}} \times \frac{6 \times 10^{23}\ \ \text{electrons}}{1\ \text{mol}} \approx f \times 10^5\ \text{J/mol} = f \times 10^2\ \text{kJ/mol}$$

MW of CH_2O is 30 Da, so $\frac{1}{30}$ mol in 1 gram.

$$\frac{300\ \text{kJ/mol}}{30\ \text{g/mol}} = 10\ \text{kJ/g} \approx 2\ \text{kcal/g}.$$

Actually 4 kcal/g as in Table 2.

8 The Challenge of Agriculture

> Humanity must now produce more food in the next four decades than we have in the last 8,000 years of agriculture combined. – Widely quoted 2012 WWF report

Who trusts this statement? **Vote**. Lets try to sanity check it.

Next four decades:
40 years of 10^{10} people: few $\times 10^{11}$ **person-years**.

Historical:

- $F0$ years of $f \times 10^9$ people: 10^{11} person-years

[3] Voltage, or electric potential difference, has dimensions of energy per charge. The energy associated with a point charge q crossing a potential difference V is $E = qV$. An electron volt is the energy associated with an electron ($q = 1.6 \times 10^{-19}$ C) crossing a 1 V potential drop, so 1 eV $= 1.6 \times 10^{-19}$ C $\times 1$ J/C $= 1.6 \times 10^{-19}$ J.

- 10^2 years of 10^9 people: 10^{11} person-years
- 10^3 years of 10^8 people: 10^{11} person-years
- 10^4 years of 10^7 people: 10^{11} person-years

Estimates of the global human population were $\approx f$ hundred million from the beginning of the Common Era through 1800 or so, when it ~ 1 billion. Remained in the 1-2 billion range til 1930 or so. Now at 8 billion.

Verdict: **passes** the sanity check.

9 Human energy production

What is the characteristic $\underbrace{\text{energy production}}_{\text{power output}}$ of a human?

What am I looking for? What would be the units of the answer? **Call out some units related to energy.** Divide into units that describe energy amount vs. power or the sort.

Some examples: Cal, kcal, W, kilowatt-hour, joule, electron volt, BTU, etc. Detail in Table 2.

Unit	Type	Conversion
cal	energy	$\simeq 4.2$ J
kcal	energy	$\simeq 4.2 \times 10^3$ J
kWh	energy	3.6×10^6 J
joule (J)	energy	1 J
eV	energy	1.6×10^{-19} J
BTU	energy	$\simeq 10^3$ J
Watt (W)	power	1 W = 1 J/s
kW	power	10^3 W
Horsepower	power	$\simeq 750$ W

Table 2: Units of energy and power. Some conversions are rounded as indicated.

What units will we use to express our answer? **Units of power** – energy over time (e.g., 1 W = 1 J/s). **10 min break here to work on it.**

How much do you eat? About 2000 kCal/day/person.

Before we answer the question, lets play with these numbers a bit

$$\approx 2000 \text{ kcal/day/person} \times \frac{1 \text{ day}}{20 \text{ hrs}} \times \frac{1 \text{ person}}{100 \text{ kg}} \approx 1 \text{ kcal/kg/hr}$$

This is a useful number to carry around. For each kg in your body, you need to feed it a kcal each hr.

Returning to our question – how do we get to watts (J/s) from kCal/day? Need to turn the kCal into J and the days into seconds.

$$\approx 2000\text{–}2500 \text{ kcal/day} \times 4000 \text{ J/kcal} \approx 10^7 \text{ J/day}$$

$$10^7 \text{ J/day} \times \frac{20 \text{ hr}}{1\text{day}} \times \frac{f \times 10^3\text{s}}{1 \text{ hr}} \times \frac{1 \text{ day}}{10^5 \text{ s}} \approx 100 \text{ J/s} = \mathbf{100\ W}$$

So we are each about **as powerful as an (old) lightbulb**... From this number we can go all sorts of places. E.g., if you have 10 friends over during the summer, can your AC keep up? A typical AC has a power output of 1000 W.

Note: 100 W produced at 10% efficiency or with 10:1 fossil fuel subsidies means 1 kW needed for food production – comparable to electricity consumption in developed countries.

So about **1 W/kg** (also ≈ 1 kcal/kg/hour).

This is about 10,000 times more than the Sun on a gram-versus-gram basis!

$$\left[\frac{4 \times 10^{26} \text{ W}}{2 \times 10^{30} \text{ kg}}\right]$$

Compare: horsepower, light bulbs, electricity consumption, energy consumption.

10 Land for food

How much **land area is needed to supply the food** for a single person?Take some time to think about it. *How would you go about this?*

Strategy (bottom-up): area needed $\leftarrow$ (energy needed, energy yield per area) $\leftarrow$ (solar energy per unit area + overall photosynthetic efficiency).

10.1 Estimate solar energy at Earth surface

In the end we need to get 100 W. Energy comes from the sun. How much energy do I get from the sun per m^2? You might remember it, but I certainly don't. Try to guesstimate: lower and upper bound, geometric mean.

I'm thinking about (old) light bulbs – how much light energy reaches us at the Earth surface in light bulb units? What's are good lower and upper bounds in light bulbs? **LB**: ≥ 1 light bulb **UB**: ≤ 100 light bulb
Using the geometric mean for all the reasons previously discussed $\rightarrow$ 10 light bulbs $\times 100$ W/bulb $= 1000$ W.

10.2 Solar energy calculation

Starting with 1000 W/m^2 but I lose some of it. Why? Day/night, hours, seasons, etc.

How much do I lose? About half from day/night, about half from seasons $\rightarrow 1/4$ remains.[4]

Effective solar radiation energy at the surface

$$\sim 1000 \text{ W/m}^2 \times \frac{1 \text{ day hr}}{2 \text{ total hrs}} \times \frac{1 \text{ growing month}}{2 \text{ total months}} \approx 250 \text{ W/m}^2$$

How do I get from energy input to food output? **I need to know something about the plant yield.**

10.3 Yield estimate

What do we grow? Mostly corn. But you can't eat that corn. It's for animals. Let's say I want to grow wheat on my m^2 and make energy bars.
LB: ≥ 1 bars **UB**: ≤ 100 bars

$$10 \text{ bars/m}^2 \times f \times 10^2 \text{ kcal/bar} \approx f \times 10^3 \text{ kcal/m}^2$$

10.4 Sanity check from crop yields

I took a look at rice yields. Anybody grew rice here?

Typical grain yields are a few tons per hectare, where 1 hectare is 100×100 m $= 10^4$ m^2. [5] The value for rice is ≈ 5 ton/hectare. [6]

Calculating based on rice, assuming 100% carbs:

$$\frac{5 \times 10^3 \text{ kg}}{10^4 \text{ m}^2} \times \frac{10^3 \text{ g}}{1 \text{ kg}} \times \frac{4 \text{ kcal}}{1 \text{ g}} \approx 2 \times 10^3 \text{ kcal/m}^2$$

[4] Note: if you compare surface area of a sphere ($4\pi R^2$) to the cross-section area (πR^2) you get this factor of 4 exactly.

[5] It's like a square football field.

[6] An interesting fact: potato yields are about 50 tons per hectare. The discrepancy between potatoes and grains is due to water content – potatoes are about 80% water while grains are ≈ 10%.

10.5 Energy efficiency of agriculture

So we get about 2000 kcal/m^2/yr. How does this compare to the 250 W we got earlier?

$$2000 \text{ kcal/m}^2\text{/yr} \times 4000 \text{ J/kcal} \times \frac{1\text{yr}}{\pi \times 10^7\text{s}} \approx \frac{10^7}{f \times 10^7} f \times 10^{-}1\text{W}$$

From ~ 200 W/m^2, this gives an overall efficiency of $\approx 0.3\%$.)

$$\text{Area} = \frac{100 \text{ W/person}}{0.5 \text{ W/m}^2} = 200 \text{ m}^2\text{/person}$$

10.6 Photosynthetic and agricultural efficiency factors

What are some factors that cause this gap between insolation and harvested yield?

- Sun spectrum ($\times 2$)
- Band gap ($\times 2$)
- Land cover and photoinhibition ($\times 5$)
- Energy currencies to carbon storage ($\times 2$)
- Plant respiration ($\times 2$)
- Harvestable yield index ($\times 2$)

$$\frac{1}{2^5 \times 5} = \frac{1}{160} \approx .5\%$$

Overall efficiency $\approx 0.5\% \approx 1$ W/m^2.

Accounting for food loss ($\times 2$) and animal products ($\times 2$):

$$\sim 1000 \text{ m}^2\text{/person} \quad \text{(not counting grazing area)}$$

For 10^{10} people: ~ 10 million km^2. Empirical value: 15 million km^2, again not including land used for grazing.

11 Bicycle vs. car

When accounting for the energy to produce food, which is more sustainable? Cycling or driving?

Say a 5 km ride each direction (1 L of gasoline vs. 1 h of cycling).

Car:

$$\frac{1 \text{ L}}{10 \text{ km}} \times \frac{1000 \text{ g}}{1 \text{ L}} \times \frac{10 \text{ kcal}}{\text{g}} = 1 \text{ kcal fuel/m}$$

Bicycle:

If riding the bike doubles the caloric rate, an extra 100 kcal for 10 km:

$$0.01 \text{ kcal food/m}$$

Given ~ 10 kcal fossil fuel per kcal food:

$$\rightarrow 0.1 \text{ kcal fuel/m}$$

Gasoline density ≈ 0.7 kg/L:

$$1 \text{ L} = 9000 \text{ kcal/kg} \times 0.7 = 6000 \text{ kcal} \quad (\approx f \text{ kg } CO_2)$$

Even with 6 GHG cal per food cal, the bicycle is still an order of magnitude more efficient ($\approx 0.f$ kg CO_2).

Leisurely walking:

$$\approx 100 \text{ kcal/hour}, \quad 5 \text{ km/hour} \Rightarrow \frac{100 \text{ kcal}}{5000 \text{ m}} = 0.02 \text{ kcal/m}$$

With fossil fuels: 0.2 kcal fossil fuels/m – much better than a car, somewhat worse than a bicycle.

12 The Challenge of Not Gaining Weight

skip

Someone you know gains 1 kg/year. How many calories eaten are stored in the extra body weight?

a. 20
b. 200
c. 2000

1 kg/year $= 9000$ kcal/year (take as given for now).

$$2500 \text{ kcal/day} \times 400 \text{ days/year} \approx 10^6 \text{ kcal/year} \Rightarrow \sim 1\%$$

Sanity check: 1 kg/day $\rightarrow f00$ kg/year; 1% is f kg/year. But caloric density is $f+$ times more.

Interesting – is the issue calorie balance or regulation/genetics/hormonal disorder?

> "We eat 1 million kCalories a year and gain 1–2 pounds from ∼3000 Cal mismatch."
>
> http://www.nytimes.com/2016/05/02/health/biggest-loser-weight-loss.html

13 Ending Module

skip

- TED calculation + vignette + final project ideas
- Send topic + one-paragraph proposal describing it

Teaser: What is faster in bacteria, transcription or translation?

a. Translation b. Transcription c. Roughly the same

13.1 Module 3 (Cell Energy)

How much power is consumed per cell? How many ATP-equivalents are consumed per cell?

$$\frac{100 \text{ W/human}}{10^{14} \text{ cells/human}} \to 10^{-12} \text{ W/cell} = 1 \text{ pW}$$

How much energy is in one ATP? (Note: this is out of equilibrium.)

ATP hydrolysis: $\Delta G \approx -50$ kJ/mol

Rule of thumb: 6 kJ/mol = one order of magnitude in concentration $[k_B T N_A \ln(10)]$

$$\frac{6000 \text{ J}}{6 \times 10^{23} \text{ molec}} \approx 10^{-20} \text{ J}$$

This must be invested to transport one molecule against a concentration gradient. An ATP pump can provide ~ 8 orders of magnitude.

$$\frac{50 \text{ kJ/mol}}{6 \times 10^{23} \text{ ATP/mol}} \approx 10^{-19} \text{ J/ATP}$$

$$\frac{10^{-12} \text{ J/s}}{10^{-19} \text{ J/ATP}} = 10^{7} \text{ ATP/s} \approx 10^{12} \text{ ATP/day}$$

(At 100% efficiency; remember $\sim 10^6$ ATP/s as a rule of thumb.)

14 What do cells "spend" ATP on?

skip

Major processes (open-ended discussion, then add):

- Protein synthesis
- Signal transduction
- Protein degradation
- Protein folding (by chaperones)
- Transport / membrane potential difference

Estimates by tissue:

- Brain: 50–60% Na/K ATPase
- Heart: 40–50% actomyosin ATPase
- Kidney: 40–70% Na/K ATPase
- Liver: 15–40% gluconeogenesis; 25% protein synthesis; 25% proton leak

14.1 *E. coli* estimates

E. coli estimate via ATP usage on protein synthesis

$$f \times 10^6 \text{ proteins/cell} \times f \times 10^2 \text{ AA/protein } \times 4\text{ATP/AA} \approx f \times 10^9 \text{ ATP}$$

to make proteins. If the generation time is ≈ 1 hr, then this is a power of

$$4 \times 10^9 \text{ATP/hr} \times \frac{50 \text{ kJ}}{\text{mol ATP}} \times \frac{1 \text{ mol ATP}}{6 \times 10^{23} \text{ ATP}} \times \frac{1 \text{ hr}}{3.6 \times 10^3 \text{ s}} \times \frac{10^3 \text{ J}}{\text{kJ}} \approx 10^{-13} \text{ W}$$

A relevant vignette from the book estimate's from *E. coli* O_2 consumption. They arrive at 10^{-12} W by starting with 30 mmol O_2 per gram of cell dry weight.

Since *E. coli* cells have masses of order picograms (10^{-12} g) the power density is ≈ 1000 W/kg, which is about 1000 times greater than we estimated for humans above (1 W/kg).

Another takeaway: estimating from protein synthesis accounts for perhaps 10-50% of the total energy expenditure in E. coli. So this is a major sink of ATP, but not all of it. What else do cells do with energy?

15 Protein Turnover Quiz

skip

If the average human cell spent *all* of its energy on protein synthesis (degradation for free), what would be the average turnover time?

a. Few minutes b. Few hours c. Few days d. Few weeks

≈ 4 ATP per amino acid polymerisation (ATP→AMP + 2 GTP→2 GDP): 2 for aa loading to tRNA, 1 for aminoacyl-tRNA binding to ribosome, 1 for ribosome translocation.

$$3 \times 10^9 \text{ proteins/cell} \times 300 \frac{\text{aa}}{\text{protein}} \approx 10^{12} \text{ aa/cell}$$
$$\times 4 \frac{\text{ATP}}{\text{aa} \cdot \text{turnover}} \rightarrow 4 \times 10^{12} \text{ ATP/cell/turnover}$$

$$\tau = \frac{4 \times 10^{12} \text{ ATP/cell/turnover}}{4 \times 10^6 \text{ ATP/cell/s}} \approx 10^6 \text{ s} \approx 10 \text{ days}$$

Cells cannot turn over very rapidly "on average" – but there is large variability among cell types. For HeLa cells, the average is about 1–2 days.

16 Teaser: Comparing Genomes – *H. sapiens* vs. *E. coli*

	Genome size	Gene number
a.	$\times 5000$	$\times 5000$
b.	$\times 500$	$\times 500$
c.	$\times 50$	$\times 50$
d.	$\times 5$	$\times 5$